

Question:

Model axial-mode helical antenna and study circular polarization and radiation characteristics using Scilab.

Scilab Code

```
clc;
clear;
xdel(winsid());

// Axial-mode Helical Antenna
// Parameters
f = 2.4e9;      // Frequency = 2.4 GHz
c = 3e8;        // Speed of light
lambda = c/f;   // Wavelength

N = 8;          // Number of turns
D = 0.125;      // Helix diameter (m)
C = %pi*D;      // Circumference
S = 0.125;      // Turn spacing (m)
alpha = atan(S/C); // Pitch angle

printf("Wavelength = %f m\n",lambda);
printf("Helix circumference = %f m\n",C);
printf("Pitch angle = %f degrees\n",alpha*180/%pi);

// Radiation pattern
theta = 0:0.5:180;

// Approximate axial-mode radiation pattern
// Maximum radiation along theta = 0 degrees
n = length(theta);
E = zeros(1,n);

for i = 1:n
    // Axial-mode approximation
    E(i) = abs((sin(N*%pi*S/lambda*cos(theta(i)*%pi/180))) / ...
        (N*sin(%pi*S/lambda*cos(theta(i)*%pi/180)) + 1e-6));
end
```

```

// Normalize
E = E/max(E);

// Convert to dB
EdB = 20*log10(E + 1e-6);

// Limit minimum value
for i = 1:n
    if Edb(i) < -40 then
        Edb(i) = -40;
    end
end

// Radiation pattern
scf(1);
plot(theta,EdB);
xlabel("Theta (degrees)");
ylabel("Normalized Radiation Pattern (dB)");
title("Axial-Mode Helical Antenna Radiation Pattern");
xgrid();

// Polar representation
theta_rad = theta*%pi/180;
r = E;

x = r.*sin(theta_rad);
y = r.*cos(theta_rad);

scf(2);
plot(x,y);
xlabel("Normalized Field");
ylabel("Normalized Field");
title("Polar Radiation Pattern of Helical Antenna");
xgrid();

// Circular polarization representation

```

```

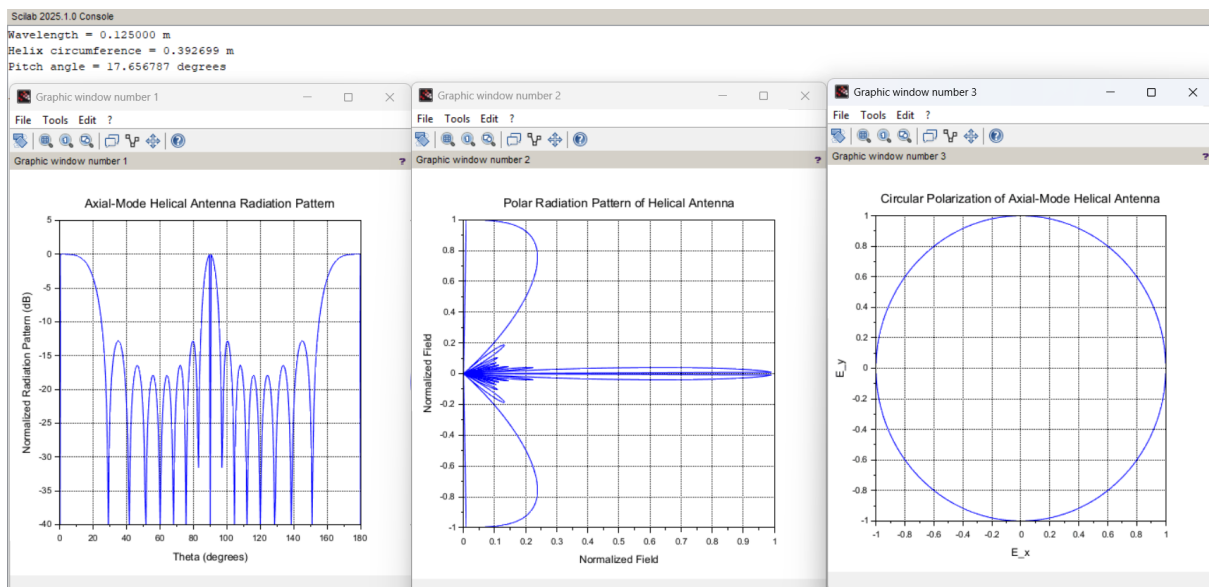
phi = 0:2*%pi/200:2*%pi;

// Right-hand circular polarization
Ex = cos(phi);
Ey = sin(phi);

scf(3);
plot(Ex,Ey);
xlabel("E_x");
ylabel("E_y");
title("Circular Polarization of Axial-Mode Helical Antenna");
xgrid();

```

OUTPUT:



Result

The Scilab simulation Radiation gives the following observations:

1. The **radiation pattern is directional**, with maximum radiation along the helix axis ($\theta = 0^\circ$).
2. The antenna operates in **axial mode**, producing a highly directional beam.
3. The electric-field components E_x and E_y have approximately equal amplitudes and a **90° phase difference**.
4. Hence, the antenna produces **circular polarization**.
5. The circular polarization plot appears approximately as a **circle**, indicating circular polarization.

6. The radiation pattern shows a strong main lobe along the antenna axis with reduced radiation at larger angles.

Result:

The axial-mode helical antenna was successfully modeled in Scilab. Its radiation characteristics were studied, and the circular polarization behavior was verified from the equal-amplitude orthogonal electric-field components having a 90° phase difference.